

Keerthi Reddy Bojja

945-296-0351 | keerthireddybojja16@gmail.com | [linkedin.com/in/keerthireddybojja](https://www.linkedin.com/in/keerthireddybojja)

Full-Stack Software Developer with 1+ years of experience building scalable web and backend systems. Skilled in Java, Python, JavaScript, Spring Boot, and cloud-native technologies. Strong understanding of software design, operating systems, networking, and testing practices.

EXPERIENCE

Software Developer

Saayam For All

May 2025 – Present

California, United States

- Architected backend **RESTful APIs** using **Java Spring Boot** and **PostgreSQL** for volunteer management platform, adhering to **Object-Oriented principles** while implementing multi-step onboarding workflows with user profile management, form validation, data persistence, and improving data processing efficiency by **20%**.
- Optimized database queries and implemented **Redis caching** for frequently accessed volunteer data, contributing to **25% improvement** in response times for search and filtering operations.
- Engineered **internationalization** support for volunteer portal using **Spring Boot MessageSource**, enabling multi-language content (Eg: English, Hindi, Chinese) and enhancing accessibility for diverse volunteer communities.
- Designed and implemented **microservices architecture** components including event publishers, message handlers, volunteer matching algorithms, and cross-service data synchronization across the **distributed system**.
- Established comprehensive unit and integration testing framework using **JUnit** and **Mockito**, achieving **80% code coverage** for assigned modules and orchestrating **CI/CD pipeline** automation with **Maven** and **GitHub Actions**.

Backend Integration Intern

UT Dallas - Center for Children and Families

Aug 2024 – Dec 2024

Texas, United States

- Collaborated with stakeholders to define system requirements and implemented backend **ETL pipeline** using **Salesforce Apex** and **HTTP callouts** to integrate external Database, automating **90%** of manual data entry and reducing processing time by **70%**.
- Enhanced dynamic search and filtering features using **React JS** and **DOM manipulation**, resulting in a **25% increase** in user engagement and improved real-time data exploration capabilities.
- Engineered multilingual UI system using **React Context API** and **i18next** in JavaScript, resulting in **30% faster** page loads and seamless language switching for **1,000+** daily users across English and Spanish interfaces.

Software Developer Intern

TCR Innovation

Sep 2022 – Jun 2023

Delhi, India

- Developed backend infrastructure for B2B, B2C, and admin platforms using **FastAPI**, improving **API response times** by **15%** through optimized routing and **asynchronous processing**.
- Tuned **MongoDB** and **MySQL** queries using **indexing** and **query refactoring**, reducing average latency by **25%** across high-traffic endpoints.
- Integrated **Google Maps API** and third-party services to power location-based features, increasing user engagement by **20%** and reducing manual location input errors by **30%**.

PROJECTS

RAG Query Decomposition System | Python, GPT-4o-mini, LangChain, FAISS

- Built advanced **RAG pipeline** using GPT-4o-mini LLM for processing **50K+ character** SEC filings with **query decomposition** for complex document analysis.
- Implemented **semantic search architecture** with **BGE embeddings**, **FAISS vector database**, and **Cohere reranking**, improving retrieval precision by **60%** through contextual compression techniques.
- Orchestrated automated evaluation framework with custom **prompt templates** across **5 performance metrics** (accuracy, coherence, relevance), achieving **92% accuracy** in document comprehension tasks.

Visual-Link: Computer Vision Product Catalog | Python, YOLOv8, OpenCV, TensorFlow

- Deployed **real-time video-based** product search platform using **YOLOv8** and **Faster R-CNN** with custom training datasets, achieving **94% object detection accuracy** for **100+ retail product categories**.
- Streamlined **ML inference pipeline** for edge deployment, reducing processing time by **15%** and enabling real-time product recognition with **sub-second response times** for mobile applications.

AI-Powered Content Generation SaaS | React, Express.js, MongoDB, OpenAI API

- Crafted **full-stack AI content platform** with responsive **React frontend**, **Express.js backend** following **MVC pattern**, and **MongoDB integration**, serving registered users with **JWT authentication** and role-based access control.
- Synchronized **OpenAI GPT models** and **Stripe payment processing** for subscription management, deployed on **AWS (EC2, S3, Lambda)** with **99.9% uptime** and automated scaling for peak traffic loads.

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, SQL

Core Concepts: Object-Oriented Programming, Design Patterns, Data Structures, Algorithms, Operating Systems

Backend & Frontend: Spring Boot, Node.js, React, HTML, CSS, PostgreSQL, MySQL, MongoDB, Redis

Cloud & Infrastructure: AWS (EC2, S3, Lambda, RDS), Docker, Kubernetes

Testing & DevOps: JUnit, Mockito, Maven, Git, GitHub Actions, CI/CD Pipelines

Certifications: AWS Certified Cloud Practitioner

EDUCATION

MS in Information Technology

University of Texas at Dallas

Aug 2023 – May 2025

Richardson, TX